Build Secure Connect

VeilChat

Secure Message System

A Software Engineering Project | Integrating 8 Virtual Lab Practicals

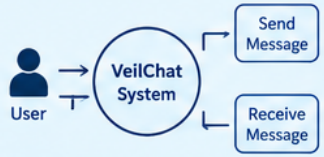
Your Messages. Your Privacy. Our Priority.

- Collected user & system requirements
- Prepared SRS document
- Defined functional & non-functional needs
- Created Gantt chart for development phases
- Estimated effort and timeline
- Planned resources and milestones

Output: SRS Document & Gantt Chart

3. Functional Modeling of the System

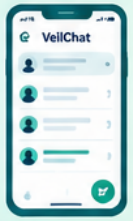
- Developed Use Case Diagrams
- Created Context Diagrams
- Designed Data Flow Diagrams (DFD)
- Prepared system context and flow



Output: Use Case, Context & DFD Diagrams

5. User Interface (UI) & Prototype Design

- Designed wireframes and UI using Figma
- Developed interactive prototype
- Focused on user-friendly and simple interface



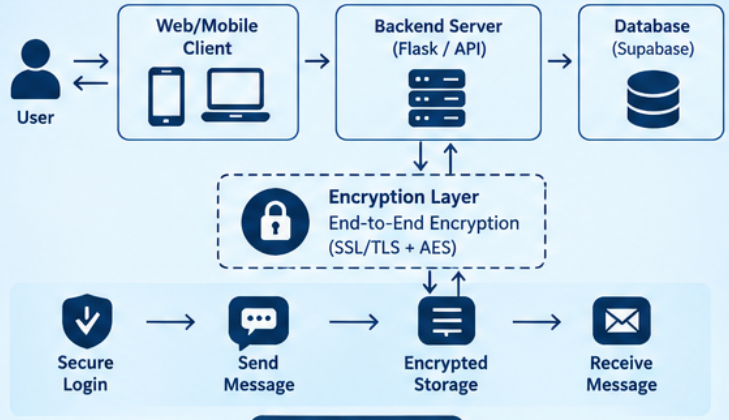
Output: UI Screens & Prototype

6. Software Testing and Test Case Design

- Wrote test cases (black-box & white-box)
- Performed functional and security testing
- Prepared Requirement Traceability Matrix (RTM)

Output: Test Cases, Test Results & RTM

VeilChat – System Architecture



Key Features

- User Authentication
- End-to-End Encryption
- Real-time Messaging
- No Forwarding / Sharing
- Cross-Platform (Web + Mobile)

7. Software Configuration Management (Git)

- Created Git repository
- Performed commit, push, pull, branch, merge
- Tracked version history



Output: Git Repository & Commit History

Project Highlights

- ✓ Secure and private communication
- ✓ User-friendly and simple interface
- ✓ Well-structured development process
- ✓ Version-controlled and well-documented
- ✓ Tested, reliable and maintainable code

From Idea to Secure Communication

8. Software Metrics & Quality Assurance

- Evaluated metrics (defect density, complexity, etc.)
- Applied quality assurance practices
- Reviewed and improved code quality

Output: Metrics Report & Quality Analysis

Outcome

- ✓ Complete system design (SRS → Deployment)
- ✓ Secure and functional messaging platform
- ✓ Version-controlled and well-documented project
- ✓ Tested, maintainable and quality-checked code

"From requirements to deployment — all 8 practicals come together to build a secure, reliable and well-engineered system."

— VeilChat

Secure Today ✦ Smarter Tomorrow